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 ****************************************************
 *                                                  *
 *           STAAD.Pro                              *
 *           Version  2006     Bld 1002.US          *
 *           Proprietary Program of                 *
 *           Research Engineers,  Intl.             *
 *           Date=    APR 28, 2007                  *
 *           Time=    15:20:39                      *
 *                                                  *
 *      USER ID: Meinhardt Infrastructure Pte Ltd   *
 ****************************************************

    1. STAAD SPACE
INPUT FILE: Electrical Store.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE APRIL 2007
    4. END JOB INFORMATION
    5. INPUT WIDTH 79
    6. UNIT METER KN
    7. JOINT COORDINATES
    8. 102 -8.000 3.982 -3.102; 104 -8.000 5.464 -2.422
    9. 110 -8.000 3.982  3.102; 502  8.000 3.982 -3.102
   10. 504  8.000 5.464 -2.422; 108 -8.000 5.464  2.422
   11. 508  8.000 5.464  2.422; 510  8.000 3.982  3.102
   13. 201  -5.000  0.000 -2.685; 202  -5.000  4.054 -3.111
   14. 203  -5.000  4.889 -2.974; 204  -5.000  5.538 -2.429
   15. 205  -5.000  5.818 -1.631; 206  -5.000  5.875  0.000
   16. 207  -5.000  5.818  1.631; 208  -5.000  5.538  2.429
   17. 209  -5.000  4.889  2.974; 210  -5.000  4.054  3.111
   18. 211  -5.000  0.000  2.685
   20. 301  0.000  0.000 -2.685; 302  0.000  4.174 -3.124
   21. 303  0.000  5.011 -2.986; 304  0.000  5.662 -2.440
   22. 305  0.000  5.943 -1.639; 306  0.000  6.000  0.000
   23. 307  0.000  5.943  1.639; 308  0.000  5.662  2.440
   24. 309  0.000  5.011  2.986; 310  0.000  4.174  3.124
   25. 311  0.000  0.000  2.685
   27. 401  5.000  0.000 -2.685; 402  5.000  4.054 -3.111
   28. 403  5.000  4.889 -2.974; 404  5.000  5.538 -2.429
   29. 405  5.000  5.818 -1.631; 406  5.000  5.875  0.000
   30. 407  5.000  5.818  1.631; 408  5.000  5.538  2.429
   31. 409  5.000  4.889  2.974; 410  5.000  4.054  3.111
   32. 411  5.000  0.000  2.685
   34. MEMBER INCIDENCES
   35. 201 201 202 210; 301 301 302 310; 401 401 402 410
   36. *
   37. 1001 104 204; 1002 204 304; 1003 304 404; 1004 404 504
   38. 2001 108 208; 2002 208 308; 2003 308 408; 2004 408 508
   39. *
   40. 3001 102 202; 3002 202 302; 3003 302 402; 3004 402 502
   41. 4001 110 210; 4002 210 310; 4003 310 410; 4004 410 510
   42. *
   43. 8001 204 308; 8002 208 304; 8003 304 408; 8004 308 404
   44. DEFINE MATERIAL START
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   45. ISOTROPIC STEEL
   46. E 2.05E+008
   47. POISSON 0.3
   48. DENSITY 76.8195
   49. ALPHA 1.2E-005
   50. DAMP 0.03
   51. END DEFINE MATERIAL
   53. MEMBER PROPERTY BRITISH
   54. 201 TO 210 301 TO 310 401 TO 410 -
   55. 1002 1003 2002 2003 TAPERED 0.3 0.006 0.3 0.15 0.008 0.15 0.008
   56. 1001 1004 2001 2004 TAPERED 0.3 0.006 0.3 0.15 0.008 0.15 0.008
   57. 3001 TO 3004 TAPERED 0.3 0.010 0.3 0.25 0.015 0.25 0.015
   58. 4001 TO 4004 TAPERED 0.3 0.010 0.3 0.25 0.015 0.25 0.015
   59. 8001 TO 8004 TABLE ST UA90X90X6
   60. CONSTANTS
   61. BETA 90 MEMB 3001 TO 3004 4001 TO 4004
   62. MATERIAL STEEL ALL
   63. MEMBER TENSION
   64. 8001 TO 8004
   65. SUPPORTS
   66. 201 211 301 311 401 411 PINNED
   67. LOAD 1 SELFWEIGHT
   68. SELFWEIGHT Y -1
   69. LOAD 2 SDL
   70. MEMBER LOAD
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   71. 1001 TO 1004 2001 TO 2004 UNI GY -3.0
   72. 3001 TO 3004 4001 TO 4004 UNI GY -2.8
   73. LOAD 3 LL
   74. MEMBER LOAD
   75. 1001 TO 1004 2001 TO 2004 UNI GY -1.5
   76. 306 CON Y -5
   77. LOAD 11 WL1 (INTERNAL SUCTION, WIND ANGLE 0)
   78. MEMBER LOAD
   79. 1001 TO 1004 2001 TO 2004 UNI GY 1.9
   80. 201 TO 203 401 TO 403 UNI GZ 3.6
   81. 301 TO 303 UNI GZ 7.2
   82. 208 TO 210 408 TO 410 UNI GZ 0.6
   83. 308 TO 310 UNI GZ 1.2
   85. LOAD 12 WL2 (INTERNAL PRESSURE, WIND ANGLE 0)
   86. MEMBER LOAD
   87. 1001 TO 1004 2001 TO 2004 UNI GY 2.6
   88. 201 TO 203 401 TO 403 UNI GZ 2.0
   89. 301 TO 303 UNI GZ 4.1
   90. 208 TO 210 408 TO 410 UNI GZ 2.2
   91. 308 TO 310 UNI GZ 4.4
   93. LOAD 13 WL3 (INTERNAL SUCTION, WIND ANGLE 90)
   94. MEMBER LOAD
   95. 1001 TO 1004 2001 TO 2004 UNI GY -1.6
   96. 201 TO 203 208 TO 210 UNI GX -1.6
   97. 401 TO 403 408 TO 410 UNI GX 1.6
   99. LOAD 14 WL4 (INTERNAL PRESSURE, WIND ANGLE 90)
  100. MEMBER LOAD
  101. 1001 TO 1004 2001 TO 2004 UNI GY 2.8
  102. 201 TO 203 208 TO 210 UNI GX -3.1
  103. 401 TO 403 408 TO 410 UNI GX 3.1
  106. LOAD 21 EQ1 (EARTHQUAKE LOAD)
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  107. JOINT LOAD
  108. 204 208 304 308 404 408 FX 12.6
  109. LOAD 22 EQ1 (EARTHQUAKE LOAD)
  110. JOINT LOAD
  111. 204 208 304 308 404 408 FZ 12.6
  113. LOAD COMB 1001 1.4DL + 1.6LL
  114. 1 1.4 2 1.4 3 1.6
  116. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL1
  117. 1 1.2 2 1.2 3 1.2 11 1.2
  118. LOAD COMB 1111 1.4DL + 1.4WL1
  119. 1 1.4 2 1.4 11 1.4
  120. LOAD COMB 1211 1.0DL + 1.4WL1
  121. 1 1.0 2 1.0 11 1.4
  122. LOAD COMB 1012 1.2DL + 1.2LL + 1.2WL2
  123. 1 1.2 2 1.2 3 1.2 12 1.2
  124. LOAD COMB 1112 1.4DL + 1.4WL2
  125. 1 1.4 2 1.4 12 1.4
  126. LOAD COMB 1212 1.0DL + 1.4WL2
  127. 1 1.0 2 1.0 12 1.4
  128. LOAD COMB 1013 1.2DL + 1.2LL + 1.2WL3
  129. 1 1.2 2 1.2 3 1.2 13 1.2
  130. LOAD COMB 1113 1.4DL + 1.4WL3
  131. 1 1.4 2 1.4 13 1.4
  132. LOAD COMB 1213 1.0DL + 1.4WL3
  133. 1 1.0 2 1.0 13 1.4
  134. LOAD COMB 1014 1.2DL + 1.2LL + 1.2WL4
  135. 1 1.2 2 1.2 3 1.2 14 1.2
  136. LOAD COMB 1114 1.4DL + 1.4WL4
  137. 1 1.4 2 1.4 14 1.4
  138. LOAD COMB 1214 1.0DL + 1.4WL4
  139. 1 1.0 2 1.0 14 1.4
  141. LOAD COMB 1021 1.2DL + 0.5LL + 1.0EQ1
  142. 1 1.2 2 1.2 3 0.5 21 1.0
  143. LOAD COMB 1022 1.2DL + 0.5LL + 1.0EQ2
  144. 1 1.2 2 1.2 3 0.5 22 1.0
  145. LOAD COMB 1121 1.2DL + 0.5LL - 1.0EQ1
  146. 1 1.2 2 1.2 3 0.5 21 -1.0
  147. LOAD COMB 1122 1.2DL + 0.5LL - 1.0EQ2
  148. 1 1.2 2 1.2 3 0.5 22 -1.0
  149. LOAD COMB 1221 0.9DL + 1.0EQ1
  150. 1 0.9 2 0.9 21 1.0
  151. LOAD COMB 1222 0.9DL + 1.0EQ2
  152. 1 0.9 2 0.9 22 1.0
  153. LOAD COMB 1321 0.9DL - 1.0EQ1
  154. 1 0.9 2 0.9 21 -1.0
  155. LOAD COMB 1322 0.9DL - 1.0EQ2
  156. 1 0.9 2 0.9 22 -1.0
  157. *SLS
  158. LOAD COMB 2001 1.0DL + 1.0LL
  159. 1 1.0 2 1.0 3 1.0
  160. LOAD COMB 2002 1.0DL + 0.5LL
  161. 1 1.0 2 1.0 3 0.5
  162. LOAD COMB 2011 DL + LL + WL1
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163. 1 1.0 2 1.0 3 1.0 11 1.0
164. LOAD COMB 2012 DL + LL + WL2
165. 1 1.0 2 1.0 3 1.0 12 1.0
   STAAD SPACE                                          -- PAGE NO.    4

166. LOAD COMB 2013 DL + LL + WL3
167. 1 1.0 2 1.0 3 1.0 13 1.0
168. LOAD COMB 2014 DL + LL + WL4
169. 1 1.0 2 1.0 3 1.0 14 1.0
171. PERFORM ANALYSIS

        P R O B L E M   S T A T I S T I C S
        -----------------------------------

   NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =    41/    50/     6
   ORIGINAL/FINAL BAND-WIDTH=    32/    11/     72 DOF
   TOTAL PRIMARY LOAD CASES =     9, TOTAL DEGREES OF FREEDOM =    228
   SIZE OF STIFFNESS MATRIX =       17 DOUBLE  KILO-WORDS
   REQRD/AVAIL. DISK SPACE  =      12.3/ 416818.9 MB

 **NOTE-Tension/Compression converged after  1 iterations, Case=    1

 **NOTE-Tension/Compression converged after  1 iterations, Case=    2

 **NOTE-Tension/Compression converged after  1 iterations, Case=    3

 **START ITERATION NO.          2
 **START ITERATION NO.          3

 **NOTE-Tension/Compression converged after  3 iterations, Case=   11

 **START ITERATION NO.          2
 **START ITERATION NO.          3

 **NOTE-Tension/Compression converged after  3 iterations, Case=   12

 **NOTE-Tension/Compression converged after  1 iterations, Case=   13

 **START ITERATION NO.          2

 **NOTE-Tension/Compression converged after  2 iterations, Case=   14

 **START ITERATION NO.          2
   STAAD SPACE                                          -- PAGE NO.    5

 **NOTE-Tension/Compression converged after  2 iterations, Case=   21

 **START ITERATION NO.          2

 **NOTE-Tension/Compression converged after  2 iterations, Case=   22

172. LOAD LIST 1001 1011 TO 1014 1021 1022 1111 TO 1114 1121 1122 -
173. 1211 TO 1214 1221 1222 1321 1322
174. PRINT SUPPORT REACTION ALL
 SUPPORT  REACTION ALL
   STAAD SPACE                                          -- PAGE NO.    6
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SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
201	1001	-0.23	77.43	-2.94	0.00	0.00	0.00
	1011	-0.20	28.34	-25.27	0.00	0.00	0.00
	1111	-0.24	20.04	-29.44	0.00	0.00	0.00
	1211	-0.17	2.17	-28.61	0.00	0.00	0.00
	1012	-0.21	23.25	-20.16	0.00	0.00	0.00
	1112	-0.25	14.10	-23.47	0.00	0.00	0.00
	1212	-0.18	-3.77	-22.64	0.00	0.00	0.00
	1013	2.81	75.56	-2.52	0.00	0.00	0.00
	1113	3.27	75.13	-2.90	0.00	0.00	0.00
	1213	3.34	57.26	-2.07	0.00	0.00	0.00
	1014	5.60	42.45	-2.23	0.00	0.00	0.00
	1114	6.53	36.51	-2.56	0.00	0.00	0.00
	1214	6.60	18.64	-1.73	0.00	0.00	0.00
	1021	-12.79	37.17	-0.75	0.00	0.00	0.00
	1022	-0.21	31.52	-15.29	0.00	0.00	0.00
	1121	12.38	79.36	-4.24	0.00	0.00	0.00
	1122	-0.20	85.01	10.31	0.00	0.00	0.00
	1221	-12.74	19.12	-0.11	0.00	0.00	0.00
	1222	-0.16	13.46	-14.66	0.00	0.00	0.00
	1321	12.43	61.30	-3.61	0.00	0.00	0.00
	1322	-0.15	66.96	10.94	0.00	0.00	0.00
211	1001	-0.23	77.24	3.03	0.00	0.00	0.00
	1011	-0.21	72.68	-10.45	0.00	0.00	0.00
	1111	-0.25	71.94	-12.31	0.00	0.00	0.00
	1211	-0.18	54.07	-13.13	0.00	0.00	0.00
	1012	-0.22	68.01	-15.67	0.00	0.00	0.00
	1112	-0.26	66.49	-18.40	0.00	0.00	0.00
	1212	-0.19	48.62	-19.22	0.00	0.00	0.00
	1013	2.81	75.41	2.58	0.00	0.00	0.00
	1113	3.27	75.13	2.90	0.00	0.00	0.00
	1213	3.34	57.26	2.07	0.00	0.00	0.00
	1014	5.60	42.31	2.29	0.00	0.00	0.00
	1114	6.53	36.51	2.56	0.00	0.00	0.00
	1214	6.60	18.64	1.73	0.00	0.00	0.00
	1021	-12.79	37.11	0.77	0.00	0.00	0.00
	1022	-0.20	84.94	-10.23	0.00	0.00	0.00
	1121	12.38	79.30	4.27	0.00	0.00	0.00
	1122	-0.21	31.47	15.27	0.00	0.00	0.00
	1221	-12.74	19.12	0.11	0.00	0.00	0.00
	1222	-0.15	66.95	-10.89	0.00	0.00	0.00
	1321	12.43	61.30	3.61	0.00	0.00	0.00
	1322	-0.16	13.48	14.61	0.00	0.00	0.00
301	1001	0.00	53.79	-1.67	0.00	0.00	0.00
	1011	0.00	18.05	-33.76	0.00	0.00	0.00
	1111	0.00	10.79	-39.51	0.00	0.00	0.00
	1211	0.00	-1.22	-38.99	0.00	0.00	0.00
	1012	0.00	14.41	-24.46	0.00	0.00	0.00
	1112	0.00	6.54	-28.66	0.00	0.00	0.00

STAAD SPACE -- PAGE NO. 7

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	1212	0.00	-5.48	-28.14	0.00	0.00	0.00
	1013	0.00	54.01	-1.71	0.00	0.00	0.00
	1113	0.00	52.75	-2.12	0.00	0.00	0.00
	1213	0.00	40.73	-1.60	0.00	0.00	0.00
	1014	0.00	35.71	-1.98	0.00	0.00	0.00
	1114	0.00	31.40	-2.43	0.00	0.00	0.00
	1214	0.00	19.38	-1.91	0.00	0.00	0.00
	1021	-12.64	39.72	-1.53	0.00	0.00	0.00
	1022	0.00	14.66	-13.80	0.00	0.00	0.00
	1121	12.64	39.71	-1.49	0.00	0.00	0.00
	1122	0.00	64.76	10.78	0.00	0.00	0.00
	1221	-12.64	27.04	-1.18	0.00	0.00	0.00
	1222	0.00	1.98	-13.45	0.00	0.00	0.00
	1321	12.64	27.03	-1.14	0.00	0.00	0.00
	1322	0.00	52.08	11.13	0.00	0.00	0.00
311	1001	0.00	55.99	1.78	0.00	0.00	0.00
	1011	0.00	57.09	-12.18	0.00	0.00	0.00
	1111	0.00	54.41	-14.18	0.00	0.00	0.00
	1211	0.00	42.39	-14.70	0.00	0.00	0.00
	1012	0.00	53.36	-21.97	0.00	0.00	0.00
	1112	0.00	50.06	-25.61	0.00	0.00	0.00